

BUSINESS INTELLIGENCE PROJECT REPORT

Australian Retail Sales Analysis Dashboard

Built with Microsoft Power BI

Financial Year Performance · Category & Chain Revenue · Manager Benchmarking · Q&A AI Feature

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Executive Summary

This report documents the design, development, and business insights from an Australian Retail Sales Analysis Dashboard built in Microsoft Power BI. The dashboard analyses 80,962 transaction records spanning January 2016 to August 2017 across two retail chains — Ready Wear and Bellings — covering 10 product categories, 8 Australian states, 21 store managers, 10 category buyers, and 98 store locations.

This project introduces several features that are new to the portfolio: a Q&A AI natural language query panel (Power BI's built-in AI feature), a star schema data model with a dedicated Dim Tables sheet containing four separate lookup dimensions (Category, Buyer, Store, Date), an Append Query to combine the main Fact Table with a supplementary August 2017 dataset, and financial year-based time intelligence using custom FY Quarter and FY Month fields.

Dashboard KPI Highlights

Metric	Value	Business Context
Current Financial Year Shown	2015/16	Dashboard defaults to first FY; slicer changes it

Metric	Value	Business Context
Cost Price Shown on Dashboard	A\$307.53K	Filtered view (2015/16) cost of goods sold
Total Units Sold (Gauge)	11M (of 22M max)	Gauge showing 11 million total units — full dataset
Count of Buyers (Q&A result)	10	Answer to 'how many buyers are there' — AI Q&A panel
Total Revenue (All Data)	A\$60,842,727	Full 80,962 rows combined Fact + Aug Data
Total Cost (All Data)	A\$34,981,314	Total cost of goods across all transactions
Total Profit (All Data)	A\$25,861,413	Revenue minus cost
Overall Profit Margin	42.5%	Healthy retail margin across all categories
Total Units Sold (All Data)	11,290,763	Across both chains and all categories
Top Revenue Category	Mens — A\$12.1M	45.9% margin — highest revenue category
Top Margin Category	Accessories — 48.9%	Highest profit margin of all 10 categories
Lowest Margin Category	Groceries — 27.0%	Commoditised — thin margins typical of groceries
Largest Chain by Revenue	Ready Wear — A\$43.1M	70.9% of total revenue
Smaller Chain	Bellings — A\$17.7M	29.1% of total revenue
Date Range	Jan 2016 – Aug 2017	20 months across FY 2015/16 and 2016/17
Financial Years Covered	2015/16 and 2016/17	Australian FY: July–June
Top Manager by Sales	Lillian Pruitt — ~38K	Highest Sum of Sale Price in manager bar chart
Tool Used	Microsoft Power BI Desktop	Multi-sheet model, Append Query, Q&A AI visual

1. Dashboard Visual Walkthrough

The dashboard uses a dark slate-grey colour scheme with white and light-grey text — a clean, professional retail analytics aesthetic. It is a single-page Power BI report with six distinct visual zones: a title/FY panel, metric cards, slicers, a bar chart by chain, a pie chart, a gauge, a Q&A AI panel, a category table, and a manager bar chart.

1.1 Zone A — Left Panel: Financial Year Title & Cost Card

ZONE A	Top-left dark panel — Financial Year label + Sum of Cost Price card
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The top-left section contains two vertically stacked elements on the dark background:

- **Financial Year Label:** Displays '2015/16' in large white bold text with 'First Financial Year' subtitle below — this is a Text Card visual linked to the selected FY slicer, showing the current filter context in human-readable form
- **Sum of Cost Price Card:** Shows 'A\$307.53K' in large orange/amber text below the FY label — this is a KPI Card visual displaying the sum of Cost Price × Total Units for the currently selected financial year and any other active filters

When the user changes the FY filter (if a slicer existed) or applies Category/Buyer slicers, both values update dynamically. The '2015/16' label and '307.53K' reflect filtered data, not the full dataset totals — confirming the dashboard is showing only the 2015/16 FY slice in the screenshot.

1.2 Zone B — Top Centre: Slicers (Category & Buyer)

ZONE B	Two dropdown slicers side-by-side — Category filter and Buyer filter
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Two Dropdown Slicers are positioned in the top centre of the dashboard:

Slicer	Field	Current State	Options
Category	Dim Tables[Category]	All (no filter applied)	Accessories, Groceries, Home, Hosiery, Intimate, Juniors, Kids, Mens, Shoes, Womens
Buyer	Dim Tables[Buyer]	All (no filter applied)	10 buyers: Amy Buchanan, Bruce Curran, Chester George, Chris Monroe, Elizabeth Gentry, Ella Hickman, Howard Wright, Maggie Mayer, Richard Carr, Sara Ferrell

Both slicers are set to 'All' in the screenshot, meaning the full filtered dataset (2015/16 FY) is visible. Selecting a specific category (e.g., 'Shoes') or buyer (e.g., 'Amy Buchanan') instantly filters every visual on the page — the bar charts, pie chart, gauge, category table, and manager chart all update simultaneously.

1.3 Zone C — Centre Left: Sum of Total Units by Chain (Bar Chart)

ZONE C	Horizontal bar chart — Total Units by Chain (Ready Wear vs Bellings)
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A Horizontal Bar Chart compares total units sold between the two retail chains:

Chain	Sum of Total Units (Displayed)	Relative Size
Ready Wear	8M	Significantly larger — ~80% of units
Bellings	2M	Smaller chain — ~20% of units

The bars are rendered in a dark teal/slate colour against the grey background. The values '8M' and '2M' are displayed as data labels directly on the bars. The x-axis likely runs to approximately 10M total.

Insight: Ready Wear is the dominant chain by unit volume — selling 4x more units than Bellings. This could reflect either more store locations, higher foot traffic, or lower average price points driving higher unit volumes.

1.4 Zone D — Centre: Cost vs Sale Price Pie Chart

ZONE D	Donut/Pie chart — Sum of Cost Price vs Sum of Sale Price split
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A Pie Chart (or Donut Chart) sits centre-stage, visualising the split between total cost and total sale price across all transactions in the current filter context:

Segment	Percentage	Colour	Meaning
Sum of Cost Price	62.62%	Light grey/silver	Total cost of goods as a share of combined (cost + sale)
Sum of Sale Price	37.38%	Dark brown/charcoal	Total revenue as a share of combined (cost + sale)

Reading this chart: The 62.62% cost slice and 37.38% sale slice don't mean cost exceeds revenue — they mean that when you add cost + sale together, cost makes up 62.62% of that combined total. This is a visual showing the relationship between cost base and revenue in absolute dollar terms, not a margin calculation.

A legend below the chart labels the two segments: 'Sum of Cost Price' (light dot) and 'Sum of Sale Price' (dark dot). The chart is set to display percentages inside each segment rather than raw values.

1.5 Zone E — Right: Q&A AI Panel & Total Units Gauge

ZONE E	Power BI Q&A natural language query box + Gauge chart for Total Units
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Q&A AI Panel — 'how many buyers are there'

The top-right corner features Power BI's Q&A (Question & Answer) AI visual — one of Power BI's most distinctive features. The user has typed the natural language query 'how many buyers are there' into the search box, and Power BI has responded with:

Q&A Element	Content	Explanation
Question typed	how many buyers are there	Natural language query entered by the user
Answer displayed	10	Count of distinct buyers in the Dim Tables[Buyer] column
Metric label	Count of Buyers	Power BI auto-generated label interpreting the query
Visual type	Card	Q&A automatically chose a Card visual for a count query

The Q&A panel demonstrates Power BI's AI capability — instead of building a new visual manually, the user simply types a question and Power BI generates the answer. This feature uses natural language processing to map question words to data columns and aggregations automatically.

Other questions that could be asked in this Q&A box:

- 'total revenue by category' → Bar chart of Sale Price by Category
- 'profit margin by chain' → Calculated measure shown as a card
- 'top 5 managers by sales' → Filtered bar chart
- 'what is the average cost price' → Single value card

Sum of Total Units Gauge

Below the Q&A panel sits a Gauge (Donut Gauge) chart showing total unit sales:

Gauge Element	Value	Configuration
Current Value	11M	SUM(Fact Table[Total Units]) for current filter context
Minimum	0M	Hard zero baseline
Maximum	22M	Set as 2× the expected value — or max possible units
Needle position	~50%	11M of 22M = 50% of the gauge arc
Colour	Dark grey arc	Matching the dashboard's dark theme

The gauge communicates that the current filtered view represents approximately half (50%) of the maximum unit capacity (22M), giving a quick sense of scale relative to the full data range.

1.6 Zone F — Bottom Left: Category Sales Table

ZONE F	Category × Sum of Sale Price data table with grand total row
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A Table visual in the bottom-left shows a breakdown of revenue (Sum of Sale Price) by product Category for the current filter context (2015/16 FY):

Category	Sum of Sale Price (2015/16 view)	Rank
Shoes	1,27,758.46	1st — top category in this FY view

Category	Sum of Sale Price (2015/16 view)	Rank
Womens	85,226.23	2nd
Juniors	73,193.93	3rd
Mens	68,680.60	4th
Kids	49,895.85	5th
Accessories	34,913.30	6th
Home	34,853.72	7th
Intimate	21,902.68	8th
Hosiery	12,360.65	9th
Groceries	6,314.19	10th — lowest revenue category
Total	5,15,099.61	Grand total for 2015/16 filtered view

Note: The values shown (e.g., 1,27,758.46) are in the Indian number format used in the dashboard (lakhs notation) — where 1,27,758 = 127,758 in Western notation. The grand total 5,15,099.61 = A\$515,099.61 for the 2015/16 filtered subset.

Shoes leads revenue in this FY slice — interesting because in the full dataset across all periods, Mens is the top category. This suggests Shoes had particularly strong 2015/16 sales relative to later periods.

1.7 Zone G — Bottom Right: Manager Sales Bar Chart

ZONE G	Vertical bar chart — Sum of Sale Price by Manager (21 managers, descending order)
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The largest visual on the dashboard — a Vertical Clustered Bar Chart — occupies the entire bottom-right section, displaying all 21 store managers ranked by their Sum of Sale Price:

Rank	Manager	Approx. Sale (2015/16 view)	State Context
1	Lillian Pruitt	~38K	NSW-based manager
2	John Gardner	~37K	Senior manager
3	Rachael Long	~37K	Near-equal to John Gardner
4	Jerry Fulton	~36K	4th by a small margin
5	Jeremy Garcia	~32K	NSW region
6	George Chester	~29K	6th
7	Terri Wright	~26K	7th
8	Brian Hubbard	~24K	8th

Rank	Manager	Approx. Sale (2015/16 view)	State Context
9	Claire Bridges	~24K	Equal 8th
10	Ray Watson	~23K	10th
11–21	Various managers	~11K–23K	Declining from Amber Ward to Michael Carney
21 (last)	Michael Carney	~11K	Lowest sales in current filter

The bars are sorted in descending order — highest-performing manager on the left, lowest on the right — creating an immediate visual ranking. The x-axis label reads 'Manager' and the y-axis shows 'Sum of Sale Price' with gridlines at 10K, 20K, 30K, and 40K intervals.

Data label insight: Lillian Pruitt leads with ~38K, but all top 4 managers are within ~2K of each other (37K–38K), suggesting a broadly competitive and balanced management team rather than one standout individual.

2. Introduction

2.1 Project Background

Retail analytics is one of the most commercially valuable applications of business intelligence. Understanding which product categories drive revenue, which chains are growing, which managers are outperforming, and how financial year performance compares to targets are questions that every retail CFO, category director, and regional manager needs answered — ideally in real time, without waiting for a weekly report from the finance team.

This Power BI project models a multi-location Australian retail business with two chains (Ready Wear and Bellings) operating across 8 states. The dashboard is designed for a Retail Director or Head of Finance who needs FY-level performance visibility sliced by category, buyer, chain, and manager — all in one interactive view.

2.2 What's New in This Project vs Previous BI Projects

Feature	This Project	Previous Projects
Q&A AI Visual	YES — natural language query panel responding to typed questions	Not used in Employee, Forecast, or Ola dashboards
Append Query	YES — Aug Data appended to Fact Table in Power Query	Previous projects used single-source or Merge only
Star Schema	YES — Dim Tables sheet contains 4 lookup dimensions in one	Employee used simple lookup; Ola used 3 separate lookup tables
Financial Year Intelligence	YES — Australian FY (July–June) with FY Qtr and FY Month	Previous projects used calendar months

Feature	This Project	Previous Projects
Indian Number Formatting	YES — Sale prices displayed in lakh notation	Not present in other projects
Multi-Manager Analysis	YES — 21 managers benchmarked in single bar chart	Employee had 10 employees; no manager ranking previously

3. Dataset Overview

3.1 Data Source

The dataset is `retail_analysis_webinar.xlsx` — a multi-sheet Excel workbook simulating an Australian retail business data model. It contains three sheets: a combined Dimension Tables sheet, the main Fact Table, and a supplementary August 2017 data file (Aug Data) that is appended to the fact table in Power Query.

Repository: github.com/princemaheta2627-glitch/retail_analysis_powerbi

3.2 Three-Sheet Architecture

Sheet	Role	Rows	Columns	Key Information
Dim Tables	Combined dimension lookup table	98 rows	15 columns	4 dimensions in one sheet: Category+Buyer, Store (Suburb+Postcode+Manager+State), and Date (Date+FY+FY Qtr+FY Month)
Fact Table	Main transaction records	76,466 rows	7 columns	Date, Chain, Postcode, Category, Total Units, Sale Price, Cost Price — Jan 2016 to Jul 2017
Aug Data	August 2017 supplement	4,496 rows	7 columns	Same schema as Fact Table — August 2017 data loaded separately and appended

Combined fact rows after Append: $76,466 + 4,496 = 80,962$ total transaction records.

3.3 Fact Table Schema

Column	Type	Description	Stats
Date	DateTime	Transaction month (monthly granularity)	Jan 2016 – Aug 2017 (20 months)
Chain	Text	Retail chain name	Ready Wear or Bellings (2 chains)
Postcode	Integer	Store postcode — FK to Dim Tables store dimension	97 unique postcodes

Column	Type	Description	Stats
Category	Text	Product category — FK to Dim Tables category dimension	10 categories
Total Units	Integer	Units sold in this transaction record	Range: 1 – 10,000+
Sale Price	Decimal	Sale price per unit in AUD	Range: A\$0.61 – A\$750.00
Cost Price	Decimal	Cost price per unit in AUD	Always $\geq$ Sale Price in raw data (Sale = revenue/unit, Cost = COGS/unit)

3.4 Dim Tables — Four Dimensions in One Sheet

The Dim Tables sheet consolidates four separate lookup dimensions into a single sheet with 15 columns. In Power Query, this sheet is split into four separate dimension tables:

Dimension	Columns Used	Unique Values	Purpose in Dashboard
Category Dim	Column 1 (Category)	10 categories	Powers Category slicer and category table
Buyer Dim	Column 2 (Buyer)	10 buyers	Powers Buyer slicer and Q&A buyer count
Store Dim	Suburb (col 4), Postcode (col 5), Manager (col 6), State (col 8)	98 stores, 21 managers, 8 states	Powers manager bar chart, state-level analysis
Date Dim	Date (col 12), Financial Year (col 13), FY Qtr (col 14), FY Month (col 15)	20 months, 2 FYs, 7 quarters	Powers FY card, time filtering, quarterly slicing

3.5 Full Dataset Statistics

Metric	Value
Total Transactions	80,962 rows (76,466 Fact + 4,496 Aug)
Date Range	January 2016 – August 2017 (20 months)
Financial Years	2015/16 (Jan–Jun 2016) and 2016/17 (Jul 2016–Jun 2017) + 2017/18 partial
Retail Chains	2: Ready Wear (70.9% revenue), Bellings (29.1% revenue)
Categories	10: Mens, Shoes, Juniors, Home, Kids, Womens, Accessories, Intimate, Groceries, Hosiery

Metric	Value
Category Buyers	10 (one buyer per category)
Store Managers	21 managers across all locations
Australian States	8: ACT, NSW, NT, QLD, SA, TAS, VIC, WA
Store Locations	98 unique postcodes
Total Revenue	A\$60,842,727
Total Cost	A\$34,981,314
Total Profit	A\$25,861,413
Overall Margin	42.5%
Total Units Sold	11,290,763

4. Power BI Data Model & Power Query

4.1 Append Query — Combining Fact + Aug Data

The first and most important Power Query step is appending the Aug Data sheet to the Fact Table to create a unified transaction table. This is done using 'Append Queries as New':

1. Open Power Query Editor in Power BI Desktop
2. Select Fact Table query in the left pane
3. Home tab → Append Queries → Append Queries as New
4. Select 'Two tables': Primary = Fact Table, Secondary = Aug Data
5. Click OK → a new query 'Append1' is created with 80,962 combined rows
6. Rename Append1 to 'All Transactions' for clarity
7. Verify: row count = 76,466 + 4,496 = 80,962 ✓
8. Close & Apply

Why Append instead of Merge? Append stacks two tables with identical schemas vertically (UNION in SQL terms). The Aug Data has the same 7 columns as the Fact Table, so appending adds the August 2017 rows to the existing dataset without any join key needed.

4.2 Dim Table Splitting in Power Query

The Dim Tables sheet contains four conceptually separate dimensions packed into 15 columns. In Power Query, reference queries are created to extract each dimension:

Dim Query	Steps in Power Query	Output
Category Dim	Reference Dim Tables → Keep only col 1 → Remove blanks → Remove duplicates	10-row Category lookup table

Dim Query	Steps in Power Query	Output
Buyer Dim	Reference Dim Tables → Keep col 2 → Remove blanks → Remove duplicates	10-row Buyer lookup table
Store Dim	Reference Dim Tables → Keep cols 4,5,6,8 (Suburb, Postcode, Manager, State) → Remove nulls	98-row Store dimension
Date Dim	Reference Dim Tables → Keep cols 12–15 (Date, FY, FY Qtr, FY Month) → Remove duplicates	20-row Date dimension

4.3 Star Schema Relationships

Relationship	From (Many)	To (One)	Join Key	Purpose
Fact → Date	All Transactions[Date]	Date Dim[Date]	Date (monthly)	Enables FY, Quarter, Month filtering
Fact → Category	All Transactions[Category]	Category Dim[Category]	Category name	Powers Category slicer
Fact → Store	All Transactions[Postcode]	Store Dim[Postcode]	Postcode	Enables State, Manager, Suburb filtering
Store → State (implicit)	Store Dim[State]	—	—	State used as hierarchy within Store Dim
Category → Buyer (implicit)	Category Dim[Category]	Buyer Dim[Buyer]	Via Dim Tables alignment	One buyer per category by design

4.4 Key DAX Measures

Revenue & Profit Measures

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- Total Revenue = SUMX('All Transactions',
-     'All Transactions'[Sale Price] * 'All Transactions'[Total Units])
-
- Total Cost = SUMX('All Transactions',
-     'All Transactions'[Cost Price] * 'All Transactions'[Total Units])
-
- Gross Profit = [Total Revenue] - [Total Cost]
-
- Profit Margin % = DIVIDE([Gross Profit], [Total Revenue], 0) * 100

```

Count of Buyers (for Q&A)

— Count of Buyers = DISTINCTCOUNT('Buyer Dim'[Buyer])

This measure is what the Q&A AI visual uses when the user types 'how many buyers are there'
— Power BI maps the question to the DISTINCTCOUNT measure and displays 10.

5. Key Findings & Business Insights

5.1 Category Revenue & Profitability

Rank	Category	Total Revenue	Profit	Margin %	Strategic Position
1	Mens	A\$12,141,189	A\$5,576,186	45.9%	Highest revenue + strong margin — core category
2	Shoes	A\$9,777,405	A\$3,960,697	40.5%	High revenue, slightly lower margin — competitive pricing
3	Juniors	A\$8,160,703	A\$3,732,725	45.7%	Growing segment with near-Mens margin
4	Home	A\$8,050,440	A\$2,715,565	33.7%	High revenue but lowest margin after Groceries — bulky goods
5	Kids	A\$7,357,910	A\$3,181,189	43.2%	Strong volume, good margin
6	Womens	A\$5,742,230	A\$2,589,088	45.1%	Lower revenue than Mens — potential growth opportunity
7	Accessories	A\$3,631,323	A\$1,776,797	48.9%	Highest margin category — premium small items
8	Intimate	A\$2,406,979	A\$1,115,067	46.3%	Niche but high margin — loyalty purchase category
9	Groceries	A\$2,190,882	A\$591,641	27.0%	Lowest margin — commodity competition
10	Hosiery	A\$1,383,666	A\$622,458	45.0%	Lowest revenue but healthy margin

Strategic insight: Accessories (48.9% margin) and Intimate (46.3%) offer the best profitability per dollar of revenue despite being lower-volume categories. Growing Accessories volume through promotions or expanded range would directly improve overall portfolio margin. Groceries (27%) is a strategic drag — it may be carried for footfall reasons rather than profitability.

5.2 Chain Performance

Chain	Revenue	Profit	Margin %	Units	Strategic Note
Ready Wear	A\$43,139,397	A\$18,092,203	41.9%	~8M	Dominant chain — 70.9% of revenue but lower margin than Bellings
Bellings	A\$17,703,331	A\$7,769,211	43.9%	~2M	Smaller chain — 29.1% revenue but 2pp higher margin

Bellings outperforms Ready Wear on margin (43.9% vs 41.9%) despite being 4x smaller by revenue. This suggests Bellings may have a more premium product mix or stronger pricing power. Understanding what Bellings does differently could help improve Ready Wear's margin.

5.3 Manager Performance (from Dashboard)

The manager bar chart shows 21 managers ranked by Sum of Sale Price. Key observations:

- Top performer: Lillian Pruitt (~38K) — NSW-based, manages stores with high-revenue categories
- Close competition: Top 4 managers (Lillian Pruitt, John Gardner, Rachael Long, Jerry Fulton) are all within ~2K of each other — suggesting performance is driven by territory/store mix rather than individual management skill
- Bottom performer: Michael Carney (~11K) — manages NT-based stores (Darwin, Alice Springs) which are smaller markets with lower footfall
- The 3.5x gap between top and bottom manager suggests significant geographic and market-mix effects rather than pure performance differences

5.4 Financial Year Performance

Period	Monthly Avg Revenue	Key Observation
2015/16 (partial — Jan–Jun 2016)	~A\$2.6M/month	Start-of-period ramp; Shoes led in this period
2016/17 (Jul 2016–Jun 2017)	~A\$3.0M/month	Full FY — highest sustained revenue period
2017/18 partial (Jul–Aug 2017)	A\$4.4M/month avg	Aug 2017 spike to A\$6.4M — possible bulk/seasonal purchase

August 2017 anomaly: The Aug Data sheet (A\$6,367,739 in a single month) is nearly double the typical monthly revenue. This could reflect a seasonal peak, a bulk order, or year-end clearance event. The Append Query correctly captures this spike in the combined dataset.

5.5 Q&A AI — Natural Language Analytics

The Q&A panel demonstrates Power BI's AI capability with the query 'how many buyers are there' → Answer: 10. Additional queries that the model supports:

Query Example	Expected Output	Visual Type
how many buyers are there	10	Card
total revenue by category	Bar chart of revenue per category	Bar Chart
which chain has highest profit	Ready Wear + profit value	Card or Bar
show managers by sale price	All 21 managers ranked	Bar Chart
average margin by category	Table or bar of margins	Table/Bar

6. Technical Skills Demonstrated

6.1 Power Query — Append & Split

- Append Queries as New: combining Fact Table (76,466 rows) + Aug Data (4,496 rows) → 80,962 combined rows
- Reference Queries: creating 4 dimension tables from a single Dim Tables source sheet without duplicating data
- Remove Duplicates and Remove Null Rows: cleaning each extracted dimension table
- Column type management: ensuring Date columns are Date type, Postcode is Integer, Price columns are Decimal

6.2 Data Modelling — Star Schema

- Creating 4 dimension tables from one source sheet via Reference Queries
- Building Many-to-One relationships: All Transactions → Date Dim, Category Dim, Store Dim
- Understanding how the Store Dim (Postcode → Manager → State) enables hierarchical filtering
- Using Postcode as the join key between fact and store dimension — a real-world data modelling pattern

6.3 DAX Measures

- SUMX() for row-level multiplication before aggregation: Sale Price × Total Units per row
- DIVIDE() for safe margin percentage calculation
- DISTINCTCOUNT() for buyer count — mapped to Q&A natural language query
- Understanding measure context: all measures automatically filter when Category, Buyer, or Date slicers are applied

6.4 Power BI Q&A AI Visual

- Inserting the Q&A visual from the Visualizations pane

- Understanding how Q&A maps natural language to data model columns and measures
- Configuring Q&A synonyms: teaching the model that 'buyers' = Buyer Dim[Buyer] and 'revenue' = Total Revenue measure
- Q&A result pinning: saving a Q&A answer as a permanent visual on the dashboard

6.5 Dashboard Design

- Dark slate-grey theme: professional retail analytics aesthetic with high contrast white text
- Gauge chart: Sum of Total Units with 0M–22M range — showing 11M current value at 50% of gauge
- Pie chart: Cost vs Sale Price split — showing the cost/revenue composition at a glance
- Category table: clean two-column layout with grand total row for quick FY revenue audit
- Manager ranking bar chart: 21 bars sorted descending — immediate performance leaderboard
- FY label card: dynamic text card showing the current Financial Year filter in large text

7. Conclusion & Future Enhancements

7.1 Project Summary

The Australian Retail Sales Analysis Dashboard demonstrates the full production BI workflow: loading multi-sheet Excel data, appending supplementary data with Append Queries, splitting combined dimension tables into proper star schema lookups, building DAX revenue and profit measures, and presenting insights through 6 visual types including the unique Q&A AI panel.

The resulting dashboard gives a Retail Director or Finance Head complete visibility into: which categories are most profitable, which chain performs better on margin, which managers are leading or lagging, how financial year performance has trended, and the cost-to-revenue composition across the business — all filterable by Category, Buyer, and Financial Year in real time.

7.2 Future Enhancements

Enhancement	Description	Power BI Feature
State-level Map	Plot revenue by Australian state on a map visual	Shape Map with AUS state GeoJSON
YoY Comparison	Side-by-side FY 2015/16 vs 2016/17 revenue comparison	SAMEPERIODLASTYEAR + bookmarks
Margin Heatmap	Category × Chain margin matrix with colour coding	Matrix + conditional formatting
Store Drill-Through	Click a manager → open their individual store detail page	Drill-through filter page

Enhancement	Description	Power BI Feature
Buyer Scorecard	Buyer performance: revenue per category they manage	Buyer Dim relationship to financials
Trend Line Page	Monthly revenue trend Jan 2016 – Aug 2017 as time series	Line chart + Date Dim hierarchy
Budget vs Actual	Add a budget sheet and compare actual vs plan	New Budget table + variance DAX

7.3 Learning Outcomes

This project introduced the Append Query pattern — a fundamental Power BI technique for combining data from multiple files or periods into a single analysis table. The splitting of the combined Dim Tables sheet into four properly typed dimension tables was also a key skill: understanding that a dimension lookup table must have one row per unique value (no duplicates) and relate to the fact table on a clean join key. The Q&A AI visual was a new addition to the portfolio, demonstrating awareness of Power BI's AI features that are increasingly expected in modern BI roles.

8. Project Information

Project Title	Australian Retail Sales Analysis Dashboard — Power BI
Author	Prince Maheta
GitHub Repository	github.com/princemaheta2627-github/retail_analysis_powerbi
Dashboard File	retail_analysis_(ans).pdf (exported from Power BI Desktop)
Data File	retail_analysis_webinar.xlsx — 3 sheets: Dim Tables (98), Fact Table (76,466), Aug Data (4,496)
Business Scope	A\$60.8M revenue 11.3M units 10 categories 2 chains 21 managers 8 states
Financial Years	2015/16 and 2016/17 (Australian FY: July–June)
Primary Tool	Microsoft Power BI Desktop

Key Power Query Techniques	Append Queries as New, Reference Queries, Dim Table splitting
Key DAX	SUMX (revenue/cost), DIVIDE (margin %), DISTINCTCOUNT (buyers), Time intelligence via Date Dim
Unique Feature	Power BI Q&A AI natural language query panel — 'how many buyers are there' → 10
Dashboard Visuals	FY Card, Cost Card, Category Slicer, Buyer Slicer, Chain Bar, Cost/Sale Pie, Gauge, Q&A Panel, Category Table, Manager Bar
Date Completed	June 2025
Report Date	June 26, 2026

This report was prepared as part of a data science & business intelligence portfolio.
Prince Maheta | Data Science Portfolio | 2025